

Seedorf Obeng-Mireku

Software Engineer | Sydney, NSW, Australia

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SUMMARY

Computer Science student at Western Sydney University who builds interactive software with real state and real edge cases — a browser-based desktop environment, an event-sourced farm platform, and a C++ routing engine with a discrete-event simulator. Comfortable designing a data model first, enforcing architectural rules with tests, and working in code I did not write. Seeking software engineering internship and graduate roles.

EDUCATION

Western Sydney University

Expected January 2027

Bachelor of Science (Computer Science), Parramatta, NSW

- Coursework: Data Structures and Algorithms, Object-Oriented Programming, Database Design and Development, Computer Networks, Systems Analysis and Design, Statistics.
- Concentration: Systems Programming.

Western Sydney University International College

2023 – 2024

Diploma in Information and Communications Technology, Sydney, NSW

PROJECTS

S-OS — Portfolio as an Operating System

github.com/AdrahSeedorf/s-os

TypeScript · Next.js · React · Zustand · Tailwind · Vitest

- Built a browser-based desktop environment where projects are installed programs: a window manager with independent lifecycle, focus and z-ordering, drag and resize, and full keyboard operation including moving and resizing without a mouse.
- Wrote the window geometry, stack ordering, command interpreter and content validation as pure logic with no React or DOM dependency, making each unit-testable in isolation. 357 tests across the system.
- Held window state in a selector-subscribed store and applied drag as a transform, committing to state only on release — a drag re-renders one window rather than every consumer at 60fps.
- Derived a virtual filesystem, a search index, the Start menu and terminal navigation from one typed content registry, so adding a project surfaces it everywhere without new code. An ESLint rule enforces the boundary that will let a database replace the flat files.
- Server-rendered every project and a recruiter view as indexable routes with structured data and generated preview images, so a JavaScript shell is still shareable and crawlable.

ADRAH Farms — Poultry Production Platform

github.com/AdrahSeedorf/farm

TypeScript · Next.js · Prisma · PostgreSQL · Auth.js · Zod

- Designed a 30-model schema for a commercial layer farm in which flock population and stock are derived from an append-only event ledger rather than stored, so every figure is reproducible and a correction is a new event carrying a reason code and an author instead of an edit to history.
- Built role-based access control around a single server-side gate that throws rather than returning false, so a missed conditional cannot leak access, and applied site scoping inside queries rather than filtering results afterwards.
- Implemented credential authentication with Argon2id hashing, database-backed sign-in throttling, and immediate revocation despite JWT sessions by re-reading the user on every request.
- Represented money as integer minor units with apportioning that guarantees split amounts sum exactly to the original, and units of measure that throw on a cross-dimension conversion.
- Kept the domain core free of species nouns — organisation, site, production unit, animal group — so supporting a second species is configuration rather than a migration. 135 tests enforce the architectural rules.

EV Network Toolkit — Routing and Siting Engine

github.com/AdrahSeedorf/ev-charging

C++17 · CMake · Catch2 · Python · GitHub Actions

- Merged two university projects with incompatible models into one engine that routes agents across a network where they compete for limited capacity at nodes, trading money against time.
- Implemented five comparable planners, including an optimal search over composite (node, charge) state, alongside four greedy strategies, and a discrete-event simulator with per-station queues and measured rather than estimated waiting time.
- Replacing the inherited static congestion model with the event-driven one showed it had overstated waits by up to 1,500x; both engines remain selectable so the correction is reproducible.

- Built an ingestion pipeline producing networks from OpenStreetMap and OSRM, with a geometric validator that rejects edges shorter than the great-circle distance between their own endpoints — which caught two impossible inherited distances and a charger power rating being read as a capacity count.
- 74 tests running in CI on GCC and Clang across Linux and macOS, Debug and Release, with warnings treated as errors.

Library Management System — Legacy Code Repair

github.com/AdrahSeedorf/library-management

Java 17 · Maven · JUnit 5 · GitHub Actions

- Took a working-looking console application and found eleven defects by compiling and running it rather than reading it, reproducing each with concrete evidence before changing a line.
- Fixed silent data loss that destroyed records on every run, a filename case mismatch that worked only on macOS, a list mutated while being indexed, and a return path that never checked who held the book and could drive a borrow count negative.
- Pinned every defect with a regression test — 8 of 36 cases passed against the original commit, 41 of 41 pass now — and wrote one commit per fix named for the failure it caused rather than the line it changed.
- Consolidated loan rules that had been implemented twice and drifted apart, which was the direct cause of one defect, and added CI running the suite on JDK 17 and 21.

Nine-Board Tic-Tac-Toe — Game Search Agents

github.com/AdrahSeedorf/NBTicTacToe

C++17 · CMake

- Implemented negamax with alpha-beta pruning, iterative deepening under a wall-clock budget, and a one-million-entry Zobrist transposition table with mate-score adjustment, reaching 1.15M nodes/sec; and a Monte Carlo tree search agent using UCT.
- Separated rules, agents and rendering so the game state performs no I/O, which is what makes the position searchable; adding an agent is one class and one line.
- 62 checks including apply/undo round-trips over 300 random games, clean under AddressSanitizer and UndefinedBehaviorSanitizer, with a test asserting the transposition table changes a fixed-depth search's speed but never its result.

SKILLS

Languages: TypeScript, JavaScript, Java, C++, Python, SQL, HTML, CSS

Frontend: React, Next.js, Tailwind CSS, state management, accessibility (WCAG), Framer Motion

Backend: Node.js, REST APIs, Prisma, PostgreSQL, MySQL, authentication and authorisation

Testing & Tooling: Vitest, JUnit, Catch2, Git, GitHub Actions, CMake, Maven, Vercel

Practices: Software architecture, event sourcing, algorithms and search, working with legacy code, secure defaults

ADDITIONAL

Honours: Third place, Regional Coding Competition — Ashanti Region, Ghana (2022)

Languages: English, French, Akan

Interests: Reading, soccer, music, cooking